

UX Principles & Visual Foundations

This document summarizes fundamental UX principles and visual design foundations that help create clear, usable, accessible, and consistent digital interfaces. It focuses on user-centricity, hierarchy, usability, consistency, accessibility, context, typography, spacing, visual hierarchy, and color theory, followed by practical examples and their planned application to the **DigiDigi** project.

I used the Refactoring UI book by Steve Schoger, Adam Wathan as a reference. It helped me to both read the book I always wanted to start and have a reliable source for my initial research.

UX Principles

1.1 User-Centricity

User-centricity means designing a product around the needs, goals, behaviors, expectations, and limitations of its users rather than around the preferences of the designer or the technology itself. Understanding what users are trying to accomplish should influence the structure of the product, the information presented to them, and the way interactions are designed.

Example:

If beginner users do not understand technical terminology, an interface should explain information in familiar language instead of assuming expert knowledge.

Key takeaway: Design decisions should begin with the user's needs and the problem they are trying to solve.

1.2 Hierarchy

Hierarchy determines the order in which users notice and understand information. Important elements should receive greater visual emphasis, while supporting information should remain available without competing for attention. Visual hierarchy can be created using size, weight, contrast, color, spacing, and position. *Refactoring UI* describes visual hierarchy as the relative importance that interface elements appear to have. When everything competes for equal attention, an interface becomes noisy and difficult to understand.

Example:

On a product card, the product name and primary action may be visually prominent, while secondary details and metadata receive less emphasis.

Key takeaway: The interface should guide users toward the information and actions that matter most.

1.3 Usability

Usability refers to how easily and efficiently users can understand a product and complete their intended tasks. A usable interface should minimize unnecessary complexity, provide understandable actions, and avoid making users spend additional effort figuring out how the interface works.

Example:

A checkout flow should ask only for necessary information, clearly communicate the next action, and avoid unnecessary steps between selecting a product and completing the purchase.

Key takeaway: Users should be able to accomplish important tasks with as little confusion and friction as possible.

1.4 Consistency

Consistency means using similar visual patterns, terminology, components, and interaction behaviors throughout a product. Users learn interfaces through repetition. If similar elements behave differently across different screens, users are forced to repeatedly learn how the product works. Consistency also reduces unnecessary design decisions. *Refactoring UI* recommends defining systems for recurring values such as font size, font weight, line height, color, margin, padding, dimensions, and border radius instead of choosing new values every time.

Example:

If the primary action uses the same button style throughout an application, users can quickly recognize which action is most important.

Key takeaway: Reusable patterns make interfaces easier to learn, more predictable, and visually coherent.

1.5 Accessibility

Accessibility means designing products so that people with different abilities and limitations can understand, navigate, and interact with them. Accessibility considerations include readable typography, sufficient contrast, clear labels, understandable states, and avoiding communication that depends entirely on color.

Example:

An error should not be communicated only by turning an input field red. An icon or written error message can communicate the same information to users who may have difficulty distinguishing colors.

Key takeaway: Important information should remain understandable without depending on a single visual cue.

1.6 Context

Context considers the circumstances in which a user interacts with a product. The user's device, environment, experience level, available time, and immediate goal can all influence what information or interaction is appropriate.

Example:

A user researching a complex product at home on a desktop computer may want detailed

comparisons, while the same person checking the product on a phone inside a store may need a faster and more focused experience.

Key takeaway: Good UX considers not only **who** the user is, but also **where, when, why, and how** the product is being used.

Typography

Typography is not only a visual styling decision. It directly affects readability, hierarchy, information density, and the overall character of an interface.

2.1 Type Scale

Using arbitrary font sizes throughout an interface can create inconsistency and make design decisions unnecessarily difficult. A type scale provides a limited set of predefined font sizes that can be reused throughout the product. *Refactoring UI* recommends defining a practical set of sizes rather than selecting a new value whenever text is added. For interface design, the authors prefer a deliberately chosen scale because rigid mathematical ratios can create inconvenient gaps between available sizes.

Example:

Instead of randomly using:

15px / 18px / 23px / 29px / 37px

a design could define a consistent scale such as:

14px / 16px / 20px / 24px / 32px / 40px

The exact values depend on the product; the important principle is limiting the available choices.

Key takeaway: Use a defined type scale instead of choosing font sizes independently for every element.

2.2 Font Weight

Font weight can create emphasis without requiring large differences in font size. For UI work, *Refactoring UI* suggests that two main weights are often enough: a normal weight around 400–500 for most text and a heavier weight around 600–700 for important content. Very light weights can become difficult to read at small sizes.

Example:

Camera Recommendation — 700

Best suited for travel and street photography — 400

The heading receives emphasis while the supporting information remains readable.

Key takeaway: Use font weight intentionally to communicate importance rather than relying only on font size.

2.3 Readability

Typography should prioritize the reading experience. Fonts designed for interface text generally need to remain clear at relatively small sizes. The book recommends avoiding condensed fonts with small x-heights for primary UI text and suggests neutral sans-serif typefaces as a safe starting point for interface design. Alignment also affects readability. For English and similar left-to-right languages, most longer text should generally remain left-aligned rather than centered.

Key takeaway: Typography should make information effortless to read before it tries to look decorative.

2.4 Line Length

Very long lines make paragraphs harder to follow because the reader's eye must travel too far before locating the beginning of the next line. *Refactoring UI* recommends approximately **45–75 characters per line** for comfortable paragraph reading.

Example:

A long description should not automatically expand across the entire width of a desktop screen. Limiting its content width can create a much more comfortable reading experience.

Key takeaway: Text width should be designed around readability rather than simply filling the available space.

2.5 Line Height

Line height should change depending on the size and nature of the text. Smaller text generally benefits from more space between lines because readers need help locating the next line. Larger headings can use tighter line heights because their characters are already easy to distinguish.

Example:

A long 16px paragraph may need generous line spacing, while a large 48px headline can use a much tighter line height.

Key takeaway: Line height should be proportional to typography rather than using one universal value everywhere.

Spacing

Spacing establishes relationships between interface elements. It helps users understand which items belong together, which sections are separate, and where their attention should move next.

3.1 White Space

White space is not wasted space. It gives content room to breathe and can make an interface easier to scan.

Example:

Two unrelated sections should usually have noticeably more space between them than a heading and the paragraph that belongs to it.

Key takeaway: Use empty space to communicate structure rather than trying to fill every available area.

3.2 Consistent Spacing

Repeatedly choosing arbitrary margin and padding values makes interfaces harder to maintain and often creates subtle visual inconsistency. A predefined spacing system restricts designers to a smaller collection of reusable values. The book recommends building a spacing and sizing scale instead of treating every distance as a new decision. Once such a system exists, designers can choose

neighboring values from the scale rather than manually fine-tuning every measurement. This also creates greater consistency across the interface.

Key takeaway: Repeated spacing values should come from a shared system.

3.3 Grouping and Proximity

Spacing communicates relationships. Elements that belong together should generally appear closer to one another, while separate groups should have more distance between them.

Example:

A camera name and its short description could have a small gap:

Sony A6700

Compact hybrid camera for travel and video.

The gap between this information and the next unrelated section, such as detailed specifications, should be larger.

Key takeaway: Proximity can communicate structure before the user reads a single word.

3.4 Spacing Scale

A useful spacing scale does not necessarily increase in perfectly linear steps. At small sizes, a difference of only a few pixels is visually significant. At larger sizes, the same difference may be barely noticeable. *Refactoring UI* therefore recommends scales where smaller values sit relatively close together and larger values become progressively farther apart.

Example:

A practical system might contain values such as:

4 / 8 / 12 / 16 / 24 / 32 / 48 / 64

The exact scale is less important than consistently using the chosen system.

Key takeaway: Create a limited spacing scale that supports both small component spacing and larger layout spacing.

Visual Hierarchy

Visual hierarchy determines which elements users notice first, second, and third. An effective interface does not make everything visually important at the same time.

4.1 Importance

Before styling an interface, content and actions should be ranked according to their importance. *Refactoring UI* notes that pages generally contain a hierarchy of primary, secondary, and tertiary actions rather than many equally important actions.

Example:

On a recommendation page:

Primary: View Recommended Camera

Secondary: Compare Camera

Tertiary: Save for Later

These actions should not all receive identical visual emphasis.

4.2 Size

Size is one tool for creating hierarchy, but it should not carry the entire responsibility. The book warns that relying too heavily on font size can produce oversized primary content and unreadably small secondary content. Weight and color can often create emphasis more effectively.

Key takeaway: Use size intentionally, but combine it with other hierarchy tools.

4.3 Weight

Font weight, borders, shapes, and other visual properties can affect how heavy an element appears. A heavier font can emphasize text without dramatically increasing its size.

Example:

Sony A6700 — semibold
26 MP APS-C Sensor — regular

The first element naturally receives more attention even when the size difference is small.

4.4 Contrast

Higher contrast attracts attention, while lower contrast can move information into the background. Weight and contrast also interact. *Refactoring UI* explains that reducing contrast can visually balance an element that feels too heavy, while increasing weight can compensate when a low-contrast element becomes too subtle.

Example:

Primary text may use a dark neutral color while secondary metadata uses a softer neutral tone.

Key takeaway: Contrast is both a readability tool and a hierarchy tool.

5.5 De-emphasis

Making the primary element louder is not always the best way to create hierarchy. Sometimes the better solution is to make competing elements quieter. *Refactoring UI* recommends de-emphasizing secondary elements when the important element does not stand out enough.

Example:

Instead of making the selected navigation item extremely large or bright, inactive items can use lower contrast.

Key takeaway: Good hierarchy is created both by emphasizing important content and de-emphasizing supporting content.

Color Theory

Color contributes to brand identity, visual hierarchy, feedback, accessibility, and the overall personality of an interface.

5.1 Primary, Secondary / Accent, and Neutral Colors

A useful interface palette usually needs more than a handful of unrelated colors. *Refactoring UI* divides a practical palette into categories that include **greys/neutral colors, primary colors, and accent colors**. Neutral colors are heavily used for text, backgrounds, panels, and form controls.

Primary colors are typically associated with important actions, active navigation, and the overall identity of the product. The book recommends creating several lighter and darker variations rather than using a single primary value.

Accent colors can communicate additional meaning, including highlighting new features or representing semantic states such as warning, success, or destructive actions.

Key takeaway: A palette should contain functional roles, not simply a collection of attractive colors.

5.2 Contrast

Contrast helps separate information and maintain readability. Important text and actions generally require stronger contrast, while supporting information can often use softer contrast. However, reducing contrast too much can make content difficult to read. The goal is to create hierarchy without sacrificing usability.

Example:

A primary button may use a high-contrast background while secondary actions use quieter styling.

5.3 Saturation and Lightness

HSL describes colors using **hue, saturation, and lightness**, making these properties easier to reason about independently. Lightness describes how close a color is to black or white, while saturation controls the intensity of the color. Instead of using one version of a brand color everywhere, designers can create a range of shades by modifying saturation and lightness.

Example:

A strong primary shade might be used for a button, a very light version for a selected-card background, and a darker version for text.

Key takeaway: A flexible color system should contain purposeful variations rather than a single value for each color.

5.4 Accessibility

Color choices must preserve enough contrast for important information to remain readable. In some situations, reversing the relationship between foreground and background can produce a more

accessible result—for example, using dark colored text on a light tinted background rather than forcing light text onto a dark saturated color.

Key takeaway: A visually attractive palette still needs to support readable and understandable content.

5.5 Do Not Rely on Color Alone

Color should support meaning rather than being the only way meaning is communicated. This is especially important for users with color-vision deficiencies. *Refactoring UI* explicitly recommends using differences such as contrast alongside color and states that color should reinforce information that the design already communicates through another method.

Example:

Instead of communicating a successful result only through green:

✓ **Compatible with your needs**

Instead of communicating an error only through red:

△ **This camera exceeds your selected budget.**

Key takeaway: Color should reinforce information, not carry the entire message.

Key Takeaways

1. Design should begin with the user's needs, goals, and context.
2. Interfaces should make important content obvious while reducing unnecessary competition for attention.
3. Typography should use a consistent type scale and prioritize readability.
4. Font size, weight, contrast, and spacing should work together to establish hierarchy.
5. Spacing should follow a reusable system and communicate relationships between elements.
6. Color palettes should contain functional primary, neutral, and accent roles with useful shade variations.
7. Accessibility should be considered throughout the interface rather than treated as an optional final improvement.
8. Consistent systems reduce design decisions and make the overall product easier to understand and maintain.

Application to DigiDigi

These principles will guide the design of **DigiDigi**, a beginner-friendly camera discovery platform.

DigiDigi will be designed around the needs of users who may have little or no technical camera knowledge. Instead of immediately presenting large amounts of technical information, the experience will prioritize the user's goals, such as what they want to photograph, their experience level, portability needs, and budget.

Visual hierarchy will prioritize the information that is most meaningful to beginners. A recommendation should first explain *which camera is recommended and why*, followed by price, strengths, limitations, and finally detailed technical specifications.

Typography will use a clear and readable type scale with a limited number of font weights. Longer explanations will maintain comfortable line lengths and line heights.

Spacing will follow a consistent scale, with related camera information grouped together and larger gaps used to separate different categories of content.

Color will be used to support brand identity, important actions, recommendations, warnings, and comparison states. However, DigiDigi will not rely on color alone to communicate important information; icons, labels, and written messages will provide additional meaning.

Most importantly, the interface will aim to reduce the intimidation that beginners often experience when researching cameras. Technical specifications will remain available, but they will support the decision-making process rather than dominate it.